Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT202		INTELLIGENT SCIENCE	

2nd SEMESTER 2024/25 FINAL EXAMINATION

UNDERGRADUATE – YEAR 3

COMPLEXITY OF ALGORITHMS

TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This is a closed-book examination.
2. Total marks available are 100. This exam will count for 80% of the final mark.
3. Answer all questions. There is NO penalty for providing a wrong answer.
4. Only English solutions are accepted. Answer should be written in the answer booklet(s) provided.
5. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Question 1. [3+10+12=25 MARKS]

A) Compute the following expressions that are given in postfix notation. Show the procedure.
1 2 + 4 * 3 + **[3 marks]**

B) What is the asymptotic value (both upper and lower bound, i.e. in Big-Theta notation) of the expression $\sum_{i=1}^{n} \log_2 i$ as a function of n by using the big-Theta notation? **[10 marks]**

C) Consider the pseudo-codes representing two computer functions called loopA and loopB as follows:

```
loopA (array T[], integer n) {
for i from n to 2
   for j from 1 to i-1
      if T[j]>T[j+1] then
      swap T[j] and T[j+1]
      endif
   endfor
endfor
}
```

```
loopB (integer n) {
   for s from 1 to n
      for t from 1 to s
         for k from 1 to t
          a←t+k
         endfor
      endfor
   endfor
}
```

1) How many times will the swap function be performed in the best case and worst case of loopA? Justify your answer. **[5 marks]**
2) Express the time complexity of the loopB function using Big-Oh notation. Justify your answer. **[5 marks]**
3) Which program is faster? Justify your answer. **[2 marks]**

Question 2. [13 MARKS]

The worst-case running time T(N) of Merge-Sort on an input sequence of size N can be characterized by the following recurrence equation:

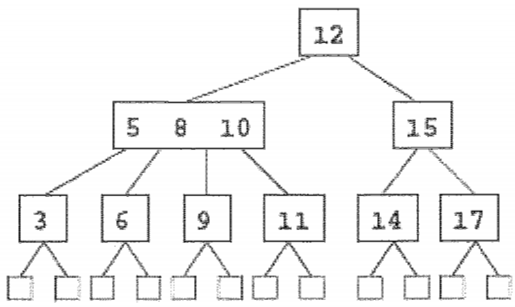
$$T(N) = \begin{cases} a & \text{if} \quad N = 1 \\ 2T(\frac{N}{2}) + bN & \text{if} \quad N > 1 \end{cases}$$

wherein a>0 and b>0 are constants.

1) Explain why the above recurrence equation can characterize the worst-case running time of Merge-Sort. **[8 marks]**
2) Solve the above recurrence equation and express the time complexity of Merge-Sort using Big-O notation. Justify your answer. **[5 marks]**

Question 3. [12 MARKS]

Let T be a (2, 4) tree shown below, which stores items with integer keys. Draw all the steps and the resulting trees obtained by performing the following operations on the original (given) T.

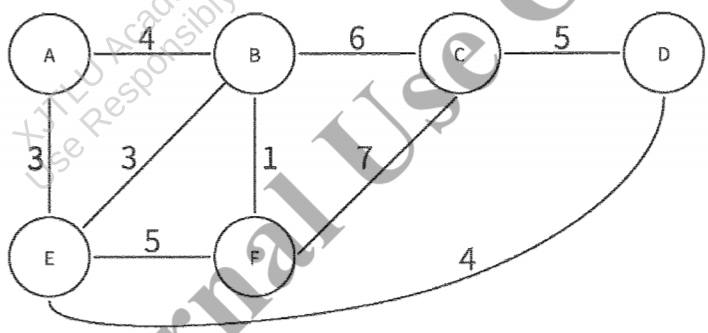


a. Insert an item with key 7 into T. **[3 marks]**

b. Remove an item with the key 17 from T. **[9 marks]**

Question 4. [10+5=15 MARKS]

Given the following weighted undirected graph:



(1) Show the step by step process of using the Kruskal's algorithm to find the minimum spanning tree. We start with node A. And draw the minimum spanning tree. **[10 marks]**

(2) What is the total weight of the minimum spanning tree? **[5 marks]**

Question 5. [7+8+5=20 MARKS]

Alice and Bob wish to use the RSA cryptosystem for communication. Bob announced the public modulus and encryption key pair to be (91, 29).
If a third-party intercepts on the message. He got the encrypted messages and the public modulus and encryption key pair to be (91, 29). Try to help him restore it to the original message.
(1) Compute Φ(n) and the prime numbers p and q associated with public modulus n. **[7 marks]**

(2) Derive the private key. **[8 marks]**

(3) What's the original unencrypted message, if encrypted message as 10? **[5 marks]**

Question 6. [5+10=15 MARKS]

A Vertex Cover set S of an undirected graph G = (V, E) is a subset of V such that every edge in E has at least one endpoint in S, if the size of S is no more than t. An Independent Set S of graph G = (V, E) is a set of vertices such that no two vertices in S are adjacent to each other. It consists of non-adjacent vertices. The Independent Set problem is to determine if the graph contains an independent set of vertices of at least size k.

(1) Is the Vertex Cover problem in NP? Prove or disprove it. **[5 marks]**

(2) Provide a reduction from Independent Set to the Vertex Cover problem, and prove it. **[10 marks]**

END OF EXAM PAPER